

$$\Delta U = W + Q$$

poveže začetno in končno stanje



$$Q=0, W=0$$

$$\Delta U=0$$

entropijski zakon

$$\Delta S \geq \int \frac{Q}{T}$$

↑  
irreverzibilne spr.  
↓  
reverzibilne spr.

diferencialni obliki

$$dU = dW + dQ$$

↓  
funk. stanja  
 $U = U(V, T)$

En. zakon za pVT sistem

$$dW = -p dV$$

$$dU = dQ - p dV$$

1)  $V = \text{konst.} : dW = 0$

$$dU = dQ = m c_v dT$$

$$c_v = \frac{1}{m} \left( \frac{\partial U}{\partial T} \right)_v$$

2)  $p = \text{konst.}$

$$dU = dQ - p dV - V dp$$

$$= dQ - d(pV)$$

$$dU + d(pV) = dQ$$

$$d(U + pV) = dQ = m c_p dT$$

$H$  (entalpija)

$$c_p = \frac{1}{m} \left( \frac{\partial H}{\partial T} \right)_p$$

Zgled: izparovanje vode

1 kg vode,  $q_i = 2,26 \text{ MJ/kg}$

$$\Delta U = ?$$

↑ povezana z entalpijo

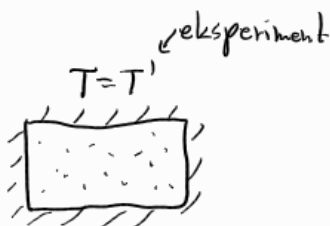
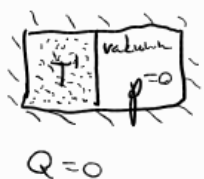
$$= Q + W$$

$$= m q_i - p(V_p - V_v) \quad \nearrow \ll V_p$$

$$\approx m q_i - p V_p = m q_i - \frac{m}{M} R T = m \left( q_i - \frac{R T}{M} \right)$$

$2,09 \text{ MJ/kg}$

## Hirnov poskus



• En. zakon:

$$\Delta U = Q + W \rightarrow Q; \text{ razpenjanje v vakuumu}$$

$\downarrow$   
 $Q; \text{ posoda je izdirana}$

$$\Delta U = 0$$

-  $U$  je funk. stanja  $U = U(T, V)$

$$\Delta U = \left( \frac{\partial U}{\partial T} \right)_V \Delta T + \left( \frac{\partial U}{\partial V} \right)_T \Delta V = 0$$

$\parallel$   
 $0$

$$\Rightarrow \left( \frac{\partial U}{\partial V} \right)_T = 0$$

$\Rightarrow U$  idealnega plina je odvisna samo od  $T$

- statistična fizika

$$U(T) = m c_v T$$

$$c_v = \frac{1}{m} \left( \frac{\partial U}{\partial T} \right)_V = c_v$$

- entalpija id. plin

$$H = U + pV = m c_v T + \frac{m}{M} R T = m \left( c_v + \frac{R}{M} \right) T$$

$$\rightarrow c_p = \frac{1}{m} \left( \frac{\partial H}{\partial T} \right)_p = c_v + \frac{R}{M}$$

$\uparrow$   
idealni plin

- razmerje spec. topl.

$$\kappa = \frac{c_p}{c_v} = \frac{c_v + \frac{R}{M}}{c_v} = 1 + \frac{R}{M c_v}$$

$\rightarrow$  st. fizika:  $c_v = \frac{3}{2} \frac{R}{M}$  za enoatomni plin

$$\kappa = \frac{5}{3}$$

$c_v = \frac{5}{2} \frac{R}{M}$  za dvoatomni plin

$\downarrow$

$$\kappa = \frac{7}{5}$$

## Zgled 1



proces hitro -  $Q=0$

sistem: plin, ki vdre v posodo

$$\Delta U = Q + W \neq 0, \text{ ker zrak, ki gre v posodo, prejme del od okolice plina}$$

$$\mu c_v (T - T') p V' = \frac{\mu R T'}{M}$$

$$\tilde{W} = -p \Delta V = -p V'$$

$$T = T' + \frac{R/M}{c_v} T' = \left( 1 + \frac{R/M}{c_v} \right) T' = \frac{c_p}{c_v} T' = \kappa T'$$

- zrak vdre v posodo

-  $T = ?$

zgleđ 2

sistem: plin v posodi + plin, ki vdre v posodo

- enačbe stanja: 
$$\left. \begin{aligned} p_1 V_1' &= \frac{m_1 R T_1'}{M} \\ p_2 V &= \frac{m_2 R T_2'}{M} \end{aligned} \right\} \text{zad. stanje}$$

$$\Delta U = W$$

$$p_1 V = \frac{(m_1 + m_2) R T}{M}$$

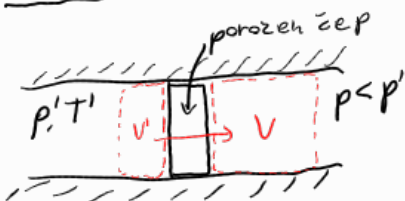
$$m_1 c_v (T - T_1') + m_2 c_v (T - T_2') = p_1 V_1'$$

$$(m_1 + m_2) T - m_1 T_1' - m_2 T_2' = \frac{m_1 R T_1'}{M c_v}$$

$$m_1 = \frac{(m_1 + m_2) T - m_2 T_2'}{T_1' \left(1 + \frac{R/M}{c_v}\right)} = \frac{\frac{p_1 V}{R/M} - \frac{p_2 V}{R/M}}{T_1' \left(1 + \frac{R/M}{c_v}\right)} = \frac{(p_1 - p_2) V}{\frac{R}{M} T_1' K}$$

$$T = \frac{p_1 V / p_1}{\frac{(p_1 - p_2) V}{K \frac{R}{M} T_1'} + \frac{p_2 V}{K \frac{R}{M} T_2'}} = \frac{K T_1'}{1 - \frac{p_2}{p_1} + K \frac{T_1'}{T_2} \frac{p_2}{p_1}} = \frac{K T_1'}{1 + \frac{p_2}{p_1} \left(K \frac{T_1'}{T_2} - 1\right)} \quad \text{-- če } p_2 = 0: \text{ isto kot prej}$$

Joule-Kelvinov pojav



topl. izolirana cev  $Q=0$   
plin

$$\Delta U = Q + W$$

0

- prejme delo od plina na levi

$$W_L = -p' \Delta V_L = -p' (0 - V') = p' V'$$

- odda delo plinu na desni

$$W_D = -p \Delta V_D = -p (V - 0) = -p V$$

$$U - U' = W = W_D + W_L = p' V' - p V$$

$$U + p V = U' + p' V'$$

$$H = H'$$

→ pri Joule-Kelvinovem procesu se  $H$  ohranja